CLASSIFICATION-
Rule Based Classifier

- Technique for classifying records using a collection of "if . . .then..." rules

Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hiber-nates	Class Label
human	warm-blooded	hair	yes	no	no	yes	no	mammal
python	cold-blooded	scales	no	no	no	no	yes	reptile
salmon	cold-blooded	scales	no	yes	no	no	no	fish
whale	warm-blooded	hair	yes	yes	no	no	no	mammal
frog	cold-blooded	none	no	semi	no	yes	yes	amphibian
komodo dragon	cold-blooded	scales	no	no	no	yes	no	reptile
bat	warm-blooded	hair	yes	no	yes	yes	yes	mammal
pigeon	warm-blooded	feathers	no	no	yes	yes	no	bird
cat	warm-blooded	fur	yes	no	no	yes	no	mammal
leopard shark	cold-blooded	scales	yes	yes	no	no	no	fish
turtle	cold-blooded	scales	no	semi	no	yes	no	reptile
penguin	warm-blooded	feathers	no	semi	no	yes	no	bird
porcupine	warm-blooded	quills	yes	no	no	yes	yes	mammal
eel	cold-blooded	scales	no	yes	no	no	no	fish
salamander	cold-blooded	none	no	semi	no	yes	yes	amphibian

r_1: (Gives Birth = no) $\wedge$ (Aerial Creature = yes) $\longrightarrow$ Birds
r_2: (Gives Birth = no) $\wedge$ (Aquatic Creature = yes) $\longrightarrow$ Fishes
r_3: (Gives Birth = yes) $\wedge$ (Body Temperature = warm-blooded) $\longrightarrow$ Mammals
r_4: (Gives Birth = no) $\wedge$ (Aerial Creature = no) $\longrightarrow$ Reptiles
r_5: (Aquatic Creature = semi) $\longrightarrow$ Amphibians

- The rules for the model are represented in a disjunctive normal form
- R is known as the rule set
- r_i 's are the classification rules or disjuncts

- Left-hand side of the rule is called the antecedent or precondition
- It contains a conjunction of attribute tests:

$$Condition_i = (A_1\ op\ v_1) \wedge (A_2\ op\ v_2) \wedge \ldots (A_k\ op\ v_k)$$

  - (A_i, v_i) is an attribute-value pair
  - op is a logical operator chosen from the set $\{=, \neq, <, >, \geq, \leq\}$
  - Each attribute test $(A_i\ op\ v_i)$ is known as a conjunct

- Right-hand side of the rule is called the rule **consequent**, which contains the predicted class y_i

- A rule r covers a record x if the precondition of r matches the attributes of x

Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hiber-nates
hawk	warm-blooded	feather	no	no	yes	yes	no
grizzly bear	warm-blooded	fur	yes	no	no	yes	yes

- r1: (Give Birth = no) $\wedge$ (Aerial Creature = yes) $\rightarrow$ Birds
- r1 covers first record but does not cover the second

- The quality of a classification rule can be evaluated using measures Coverage and Accuracy
- Given a data set D and a classification rule r : A -> y

- **Coverage:**
  - Fraction of records in D that trigger the rule r (Fraction of records that satisfy the antecedent of a rule)
- **Accuracy**
  - Fraction of records triggered by r whose class labels are equal to y (Fraction of records that satisfy the antecedent that also satisfy the consequent of a rule)

$$\text{Coverage}(r) = \frac{|A|}{|D|}$$

$$\text{Accuracy}(r) = \frac{|A \cap y|}{|A|},$$

- |A| is the number of records that satisfy the rule antecedent
- |A ∩ y | the number of records that satisfy both the antecedent and consequent
- Example:

$$(\texttt{Gives Birth} = \texttt{yes}) \wedge (\texttt{Body Temperature} = \texttt{warm-blooded}) \longrightarrow \texttt{Mammals}$$

- Coverage of 33% since five of the fifteen records support the rule antecedent
- Accuracy is 100% because all five vertebrates covered by the rule are mammals

How a Rule-Based Classifier Works

Rule based classifier classifies a test record based on the rule triggered by the record

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds
R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes
R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals
R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles
R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
lemur	warm	yes	no	no	?
turtle	cold	no	no	sometimes	?
dogfish shark	cold	yes	no	yes	?

A lemur triggers rule R3, so it is classified as a mammal
A turtle triggers both R4 and R5
A dogfish shark triggers none of the rules

Characteristics of Rule Sets

Mutually Exclusive Rules

- Rules in a rule set R are mutually exclusive if no two rules in R are triggered by the same record
- Property ensures that every record is covered by at most one rule in R

Exhaustive Rules

- A rule set -R has exhaustive coverage if there is a rule for each combination of attribute values
- This property ensures that every record is covered by at least one rule in R

If the Rule set is not exhaustive

A record may not trigger any rules

Solution?

default rule $r_d: () \longrightarrow y_d$ must be added to cover the remaining cases

- A default rule has an empty antecedent and is triggered when all other rules have failed
- y_d is known as the default class and is typically assigned to the majority class of training records not covered by the existing rules

- **If the rule set is not mutually exclusive**
- record can be covered by several rules, some of which may predict conflicting classes

- Solution?

  Ordered rule set

  Unordered rule set

Ordered Rules

- Rules in a rule set are ordered in decreasing order of their priority(based on accuracy, coverage etc)
- An ordered rule set is also known as a decision list
- When a test record is presented, it is classified by the highest-ranked rule that covers the record
- This avoids the problem of having conflicting classes predicted by multiple classification rules

Ordered Rule Set

- When a test record is presented to the classifier
  - It is assigned to the class label of the highest ranked rule it has triggered
  - If none of the rules fired, it is assigned to the default class

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds
R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes
R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals
R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles
R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
turtle	cold	no	no	sometimes	?

Unordered Rules

- Approach allows a test record to trigger multiple classification rules
- It then considers the consequent of each rule as a vote for a particular class
- Votes are then tallied to determine the class label of the test record
- The record is usually assigned to the class that receives the highest number of votes

Rule-Ordering Schemes

- Rule ordering can be implemented on a rule-by-rule basis or on a class-by-class basis

Rule-Based Ordering Scheme

- This approach orders the individual rules by some rule quality measure
- This ordering scheme ensures that every test record is classified by the "best" rule covering it
- A potential drawback of this scheme is that lower-ranked rules are much harder to interpret because they assume the negation of the rules preceding them

Class-Based Ordering Scheme

- In this approach, rules that belong to the same class appear together in the rule set R
- The rules are then collectively sorted on the basis of their class information
- The relative ordering among the rules from the same class is not important

Rule-Based Ordering

(Skin Cover=feathers, Aerial Creature=yes) ==> Birds

(Body temperature=warm-blooded, Gives Birth=yes) ==> Mammals

(Body temperature=warm-blooded, Gives Birth=no) ==> Birds

(Aquatic Creature=semi)) ==> Amphibians

(Skin Cover=scales, Aquatic Creature=no) ==> Reptiles

(Skin Cover=scales, Aquatic Creature=yes) ==> Fishes

(Skin Cover=none) ==> Amphibians

Class-Based Ordering

(Skin Cover=feathers, Aerial Creature=yes) ==> Birds

(Body temperature=warm-blooded, Gives Birth=no) ==> Birds

(Body temperature=warm-blooded, Gives Birth=yes) ==> Mammals

(Aquatic Creature=semi)) ==> Amphibians

(Skin Cover=none) ==> Amphibians

(Skin Cover=scales, Aquatic Creature=no) ==> Reptiles

(Skin Cover=scales, Aquatic Creature=yes) ==> Fishes

How to build a rule based classifier

- To build a rule-based classifier, we need to extract a set of rules that identifies key relationships between the attributes of a data set and the class label

- Methods for extracting classification rules:
  - Direct methods
  - Indirect methods

- **Direct methods**

- Extract classification rules directly from data

- Partition the attribute space into smaller subspaces so that all the records that belong to a subspace can be classified using a single classification rule

- **Sequential covering algorithm is often used to extract rules directly from data**

- Algorithm extracts the rules one class at a time for data sets that contain more than two classes

- Criterion for deciding which class should be generated first depends on a number of factors, such as the class prevalence or the cost of misclassifying records from a given class

Sequential covering algorithm

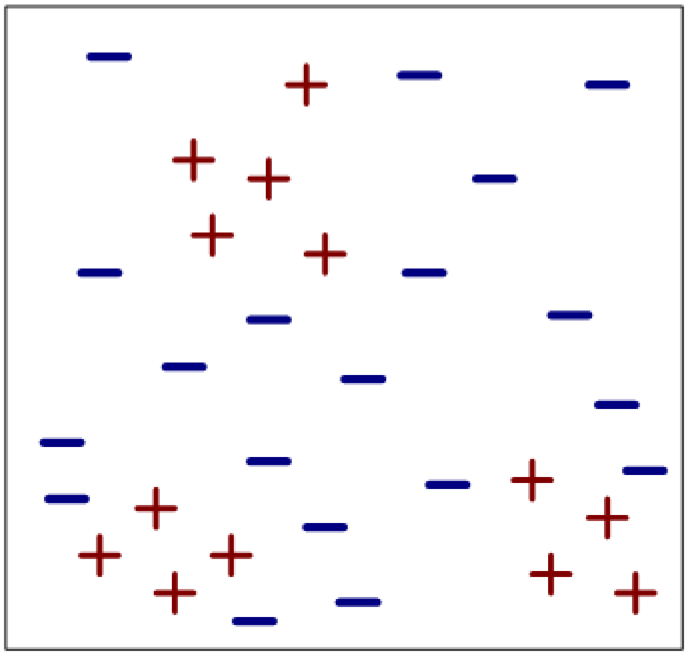
- Algorithm starts with an empty decision list R
- Learn-One-Rule function is then used to extract the best rule for class y that covers the current set of training records
- During rule extraction, all training records for class y are considered to be positive examples, while those that belong to other classes are considered to be negative examples
- A rule is desirable if it covers most of the positive examples and none (or very few) of the negative examples

- Once such a rule is found, the training records covered by the rule are eliminated
- The new rule is added to the bottom of the decision list R
- This procedure is repeated until the stopping criterion is met
- The algorithm then proceeds to generate rules for the next class

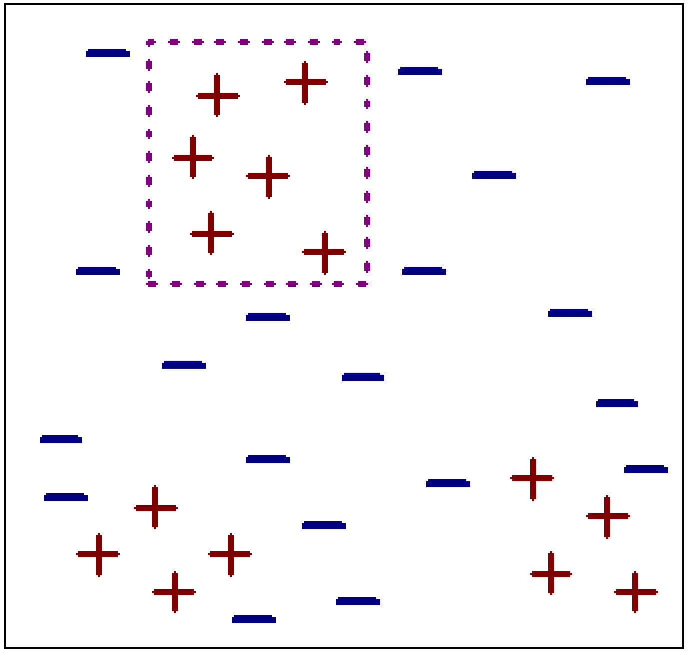
Algorithm 5.1 Sequential covering algorithm.

Let E be the training records and A be the set of attribute-value pairs, $\{(A_j, v_j)\}$.
Let Y_o be an ordered set of classes $\{y_1, y_2, \ldots, y_k\}$.
Let $R = \{\ \}$ be the initial rule list.
for each class $y \in Y_o - \{y_k\}$ **do**
    while stopping condition is not met **do**
        $r \leftarrow$ Learn-One-Rule (E, A, y).
        Remove training records from E that are covered by r.
        Add r to the bottom of the rule list: $R \longrightarrow R \vee r$.
    end while
end for
Insert the default rule, $\{\} \longrightarrow y_k$, to the bottom of the rule list R.

Example of Sequential Covering

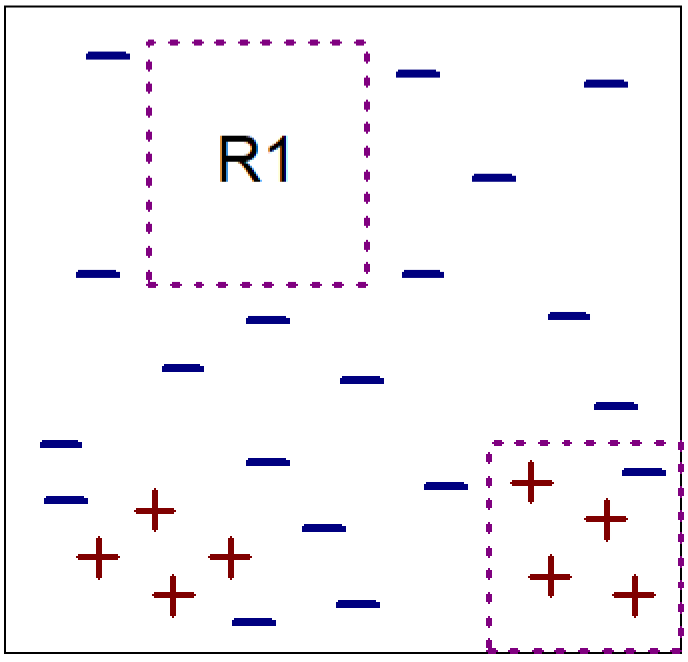


(i) Original Data

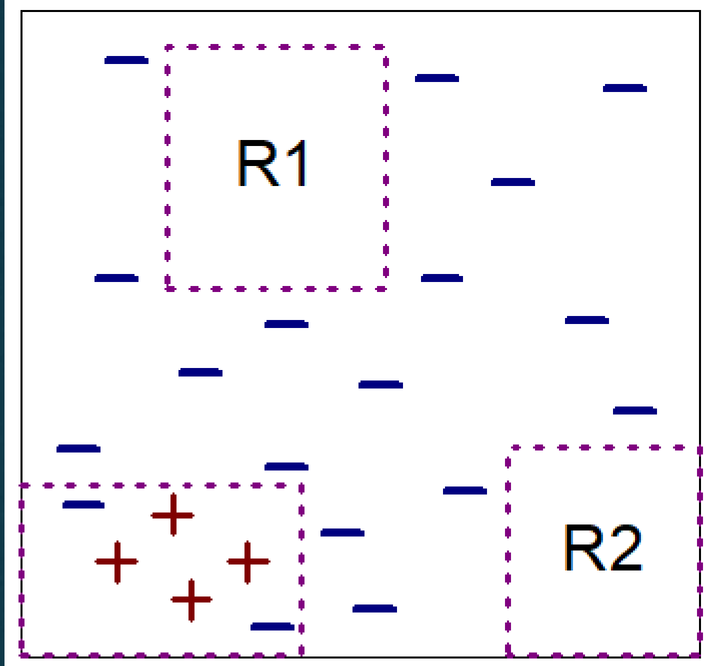


(ii) Step 1

Example of Sequential Covering



(iii) Step 2



(iv) Step 3

- Rule R1, whose coverage is extracted first because it covers the largest fraction of positive example

- All the training records covered by R1 are subsequently removed and the algorithm proceeds to Iook for the next best rule which is R2

Learn-one-Rule function

- Objective is to extract a classification rule that covers many of the positive examples and none (or very few) of the negative examples in the training set
- However, finding an optimal rule is computationally expensive given the exponential size of the search space

Learn-one-Rule function

- Function addresses the exponential search problem by growing the rules in a greedy fashion
- It generates an initial rule r and keeps refining the rule until a certain stopping criterion is met
- The rule is then pruned to improve its generalization error

Rule-Growing Strategy

- Common strategies for growing a classification rule:
  - General-to-specific
  - Specific-to-general

- General-to-specific strategy
  - An initial rule r: { } → y is created, where the left-hand side is an empty set and the right-hand side contains the target class
  - Rule has poor quality because it covers all the examples in the training set
  - New conjuncts are subsequently added to improve the rule's quality

- Specific-to-general strategy
  - One of the positive examples is randomly chosen as the initial seed for the rule-growing process
  - During the refinement step, the rule is generalized by removing one of its conjuncts so that it can cover more positive examples

Characteristics of rule based classifier

- Expressiveness of a rule set is almost equivalent to that of a decision tree
- Generally used to produce descriptive models that are easier to interpret, but gives comparable performance to the decision tree classifier
- The class based ordering approach adopted by many rule based classifiers is well suited for handling data sets with imbalanced class distributions